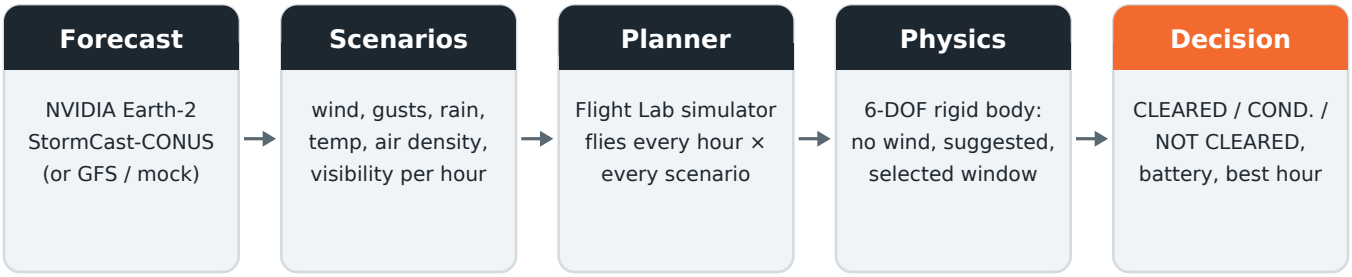
Drone Scout

Flight Lab + Mission Planner · pre-flight decisions for crop-survey drones

What the project does

A grower or researcher wants to survey a field with a drone in two or three days. Before they drive out, Drone Scout answers: **will the drone finish the survey safely at that time, how much battery will it take, and is there a better hour?** It forecasts the weather at the field, flies the real survey route in a simulator in every forecast scenario, and turns the result into a flight-release status (CLEARED, CONDITIONAL or NOT CLEARED) with reasons.



What is in it

- **Flight Lab**: a quadcopter simulator (DJI Phantom 4 Multispectral reference) flying lawn-mower survey lanes over real UF/IFAS-supplied field imagery, with wind, gusts, GNSS noise, battery and camera triggers.
- **Weather**: NVIDIA Earth-2 **StormCast-CONUS** (AI, 3 km, hourly, added to Earth2Studio in July 2026) steered by NOAA GFS for days-ahead plans; GFS alone without a GPU; a mock forecast for demos.
- **Mission planner**: flies every launch hour × every forecast scenario, finds the best hour and the battery needed.
- **6-DOF physics**: a rigid-body model (attitude, four motors, flight controller, Dryden turbulence, RK4 at 500 Hz) that shows how wind pushes the drone off its path and how it tilts and rotates, checked against hand calculations.
- **Dust and visibility**: blowing dust and rain lower visibility; under the FAA 3-statute-mile minimum (14 CFR 107.51(c)) the hour is NOT CLEARED.

How to read the map

- **Green**: original path with no wind
- **Orange**: recommended path (best launch hour, typical scenario)
- **Blue**: the window you selected, and the live simulator flight

Every planner and physics view shows these three paths, so you see at a glance how your window compares with calm air and with the best hour.

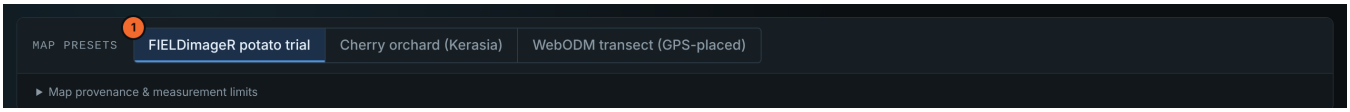
Run it

```
cd dronesim && python -m http.server 8000 # then open http://127.0.0.1:8000/index.html
# or double-click index.html and use Load forecast (button 8) · tests: node
tests/physics.test.cjs
```

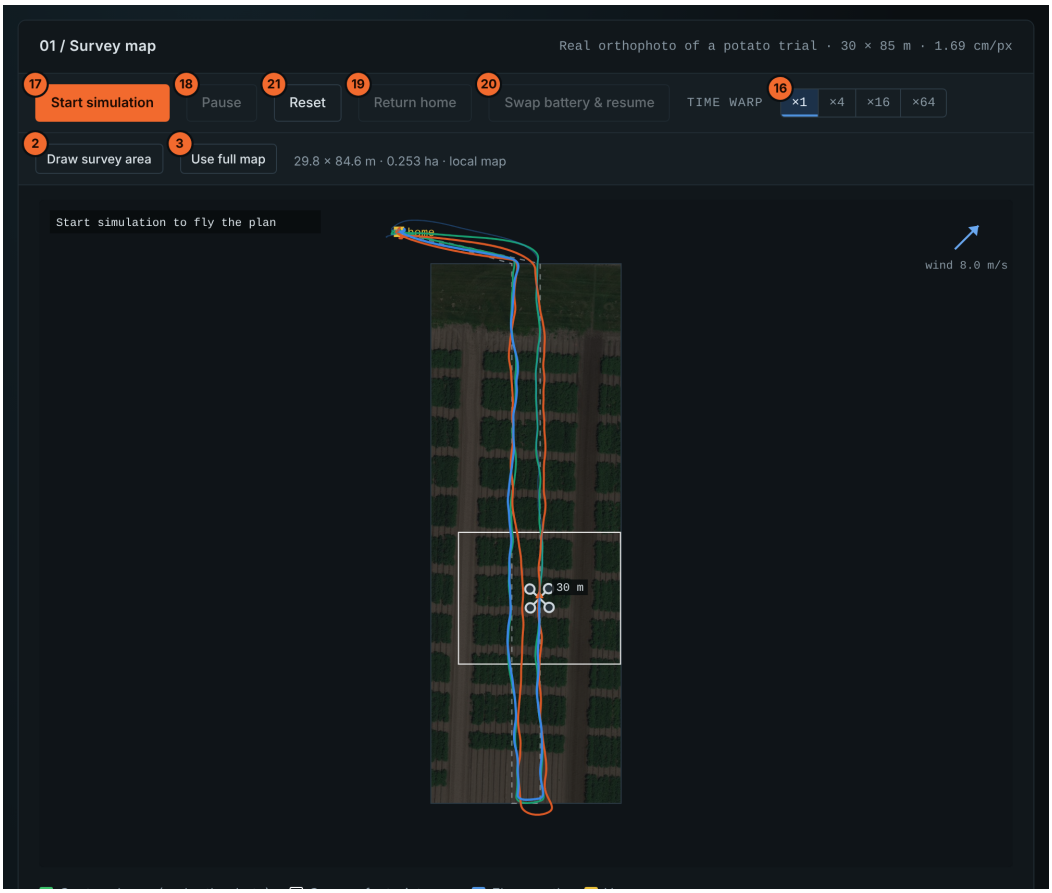
Every button, in the order you use it

The orange numbers on the screenshots match the steps. A normal run is steps 1 → 3, then 8 → 15, then 22 → 27. Steps 4–7 are only needed to change the aircraft; 16–21 control a live flight; 29–32 export a live flight.

Stage A · Choose the field

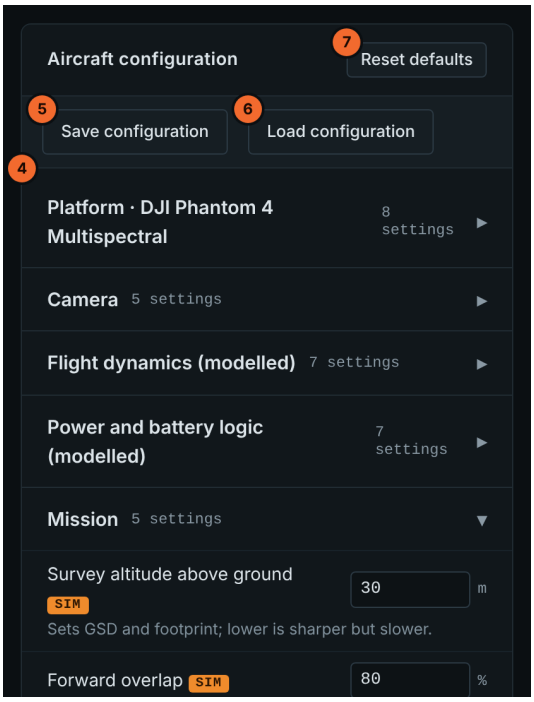


#	Button	What it does	When
1	**Map presets**	Pick the field: potato trial, cherry orchard or WebODM transect.	First, every session



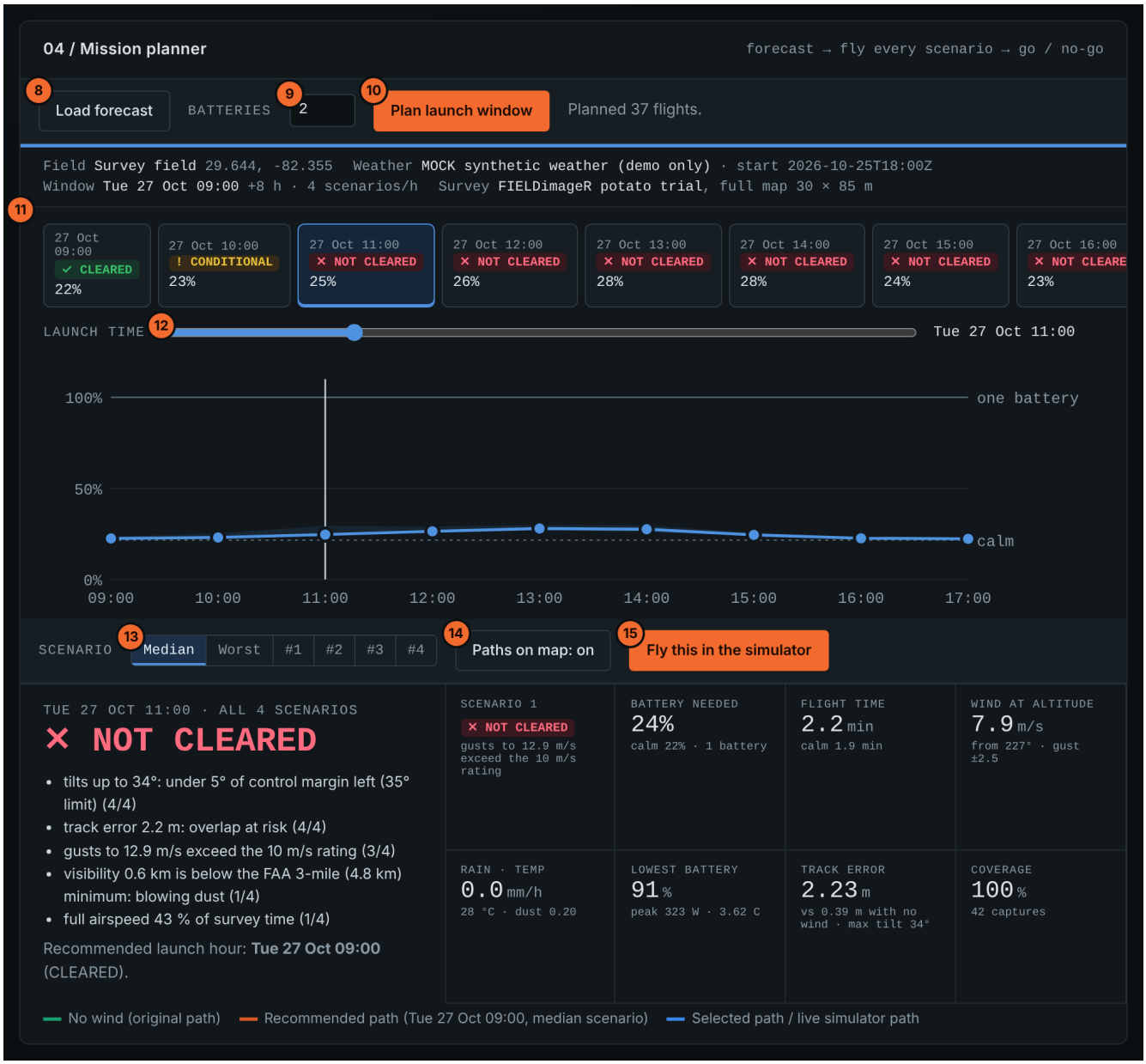
#	Button	What it does	When
2	**Draw survey area**	Drag a rectangle on the map to survey only part of the field.	When you want a portion
3	**Use full map**	Go back to surveying the whole field.	To undo step 2

Stage B · Aircraft (optional)



#	Button	What it does	When
4	**Aircraft settings**	Altitude, overlap, speeds, battery, reserve, wind sensors. Tags show SPEC / DERIVED / ASSUMED / SIM.	Only to change the drone or survey
5	**Save configuration**	Download the current settings as JSON.	Before experimenting
6	**Load configuration**	Load a saved JSON (also the replay files from the planner).	To restore or replay
7	**Reset defaults**	Back to the P4 Multispectral reference values.	If settings get messy

Stage C · Plan the launch window (04 / Mission planner)



#	Button	What it does	When
8	**Load forecast**	Pick forecast_scenarios.json (loads by itself when the page is served).	Only if opened as a file
9	**Batteries**	How many batteries you will carry.	Before planning
10	**Plan launch window**	Flies the survey in every hour × scenario with your settings and drawn area.	Main step
11	**Hour tiles**	Each launch hour with its decision and battery needed. Click one to select it.	To compare hours
12	**Launch time slider**	Same as the tiles; also ← / → keys. The map redraws the blue path.	To scan the day

#	Button	What it does	When
13	**Scenario**	Median, Worst, or one forecast scenario (#1…#n) for that hour.	To see the spread
14	**Paths on map**	Show or hide the green / orange / blue paths on the survey map.	For a clean map
15	**Fly this in the simulator**	Puts that hour’s weather into the aircraft and flies it live in Flight Lab.	To watch the flight

Stage D · Watch a live flight (01 / Survey map)

#	Button	What it does	When
16	**Time warp ×1 ×4 ×16 ×64**	Playback speed of the live flight.	Use ×16–×64 for demos
17	**Start simulation**	Fly the current plan with the current settings.	After 15, or on its own
18	**Pause / Resume**	Freeze and continue the live flight.	To explain a moment
19	**Return home**	Abort the survey and fly home.	To show a manual abort
20	**Swap battery & resume**	After a battery return, continue with a fresh pack.	Big fields
21	**Reset**	Clear the live flight.	Before a new run

Stage E · Physics (05 / Physics 6-DOF)



#	Button	What it does	When
22	**Run 6-DOF physics**	Flies the survey as a rigid body three times: no wind, recommended path, selected window.	After choosing an hour
23	**Play / Pause**	Plays the physics flight on the map and the 3D aircraft view (roll, pitch, heading).	To show motion
24	**Speed ×1 ×10 ×60**	Playback speed.	
25	**Scrubber**	Jump to any moment; the charts' cursor follows. You can also click a chart.	To find gust hits
26	**Show on map**	Show or hide the physics paths and drone on the map.	

#	Button	What it does	When
27	**Download CSV (MATLAB)**	All three flights, 20 samples/s: position, roll/pitch/yaw, rates, motor thrust, power, wind, off-path distance.	For MATLAB: plot_physics(file)
28	**Model, equations…**	Equations, controller, airframe values with their basis.	For technical judges

Stage F · Export a live flight (03 / Survey captures)



#	Button	What it does	When
29	**Export mission**	Full mission JSON: telemetry, captures, configuration, model notes.	After a live flight
30	**Download CSV**	One row per camera capture.	
31	**Footprints JSON**	Capture footprints in local map metres.	
32	**Show text to copy**	The same data as text, if downloads are blocked.	

How the decision is made

Each forecast scenario gets a status. An hour is NOT CLEARED if half or more of its scenarios are, CONDITIONAL if any scenario is NOT CLEARED or half are CONDITIONAL, otherwise CLEARED. The recommended path is the typical (median) scenario of the best hour.

Check	NOT CLEARED when	CONDITIONAL when	Source
Wind and gusts	mean + 2 × gust σ above 10 m/s	over 5 % of survey time at full airspeed; tilt over 30° (under 5° of margin)	Phantom 4-family 10 m/s; simulator
Rain	over 0.5 mm/h	some scenarios show showers	radar reflectivity → rain rate
Visibility (dust, rain)	under 4.8 km (3 statute miles)	under 8 km: keep line of sight, hazy images	FAA 14 CFR 107.51(c); estimate
Temperature	outside 0–40 °C	under 10 °C: warm the batteries	DJI P4M spec
Battery	more batteries needed than carried; emergency landing	uses every battery; lands near the reserve	simulator
Tracking	—	track error over 2 m (photo overlap at risk); coverage under 95 %	simulator

Why dust matters

Dust barely changes how the drone flies, but it can make flying illegal and the images useless. Blowing dust and rain both scatter light, so their effects add. When estimated visibility falls under the FAA minimum of 3 statute miles from the control station, the hour is NOT CLEARED even if the drone could physically fly it. In the sample forecast the 11:00 flight stays within 1 m of its path in the physics model, yet the hour is NOT CLEARED because of dust visibility.

What is measured, derived and assumed

Piece	Basis
Wind, temperature, pressure, radar reflectivity	Weather model output (StormCast-CONUS or GFS)
Wind at flight altitude, gust size	Assumed profile (exponent 0.14) and turbulence intensity 0.15, plus ensemble spread
Rain rate, air density	Derived with standard formulas (Marshall–Palmer; ideal gas)
Visibility from dust and rain	Assumed empirical relation: an estimate, not a model output
Hover power vs air density, cold batteries	Derived (momentum theory) and assumed (−0.75 %/°C below 20 °C)
6-DOF airframe: inertia, drag, motor lag, thrust-to-weight	Assumed; mass, size and propellers follow the P4 Multispectral
Physics accuracy	Hover thrust = mg/4, hover power = 177 W, steady-wind tilt = atan(drag/weight) to 0.01°, power ∝ 1/√ρ

Technical basis

Why this drone

The aircraft is the **DJI Phantom 4 Multispectral (P4M)**: a purpose-built agricultural survey drone with one RGB and five narrow-band multispectral cameras, widely used for crop-health (NDVI) mapping. It matches the multispectral field imagery in the project, and DJI publishes a full spec sheet, so the aircraft limits come from the manufacturer rather than from guesses. Every value can be changed in the Aircraft configuration panel to model another drone.

Drone and battery model

Item	Value used	Basis
Takeoff mass, size	1.487 kg, 350 mm diagonal	DJI spec
Speeds	14 m/s max, 6 m/s climb, 3 m/s descent	DJI spec (50 km/h P-mode)
Battery	Intelligent Flight Battery, LiPo 4S, 5870 mAh, 15.2 V = **89.2 Wh**	DJI spec
Endurance check	hover 165 W + 12 W payload: about 28 min hover from 100 %	Assumed; DJI states ~27 min
Power in flight	$P = 12 + 165 \cdot (1 + 0.004 \cdot v^2)$ W + climb power $m \cdot g \cdot v_z / 0.85$	Assumed empirical model (Flight Lab)
Power in 6-DOF	per rotor $P \propto T^{1.5} / \sqrt{(2 \rho A)}$, calibrated to the same 177 W hover	Momentum theory
Battery logic	return home at 15 % plus energy to reach home; land at 10 %; swap batteries if carried	Assumed operating practice
Weather on battery	hover power × $\sqrt{(1.225/\rho)}$; capacity −0.75 %/°C below 20 °C (floor 80 %)	Derived / assumed
Not modelled	voltage sag, internal resistance, battery heating and ageing, motor efficiency maps	Known limit

Mission limits and why

Limit	Value	Why
Wind incl. gusts	mean + 2σ ≤ 10 m/s	Phantom 4-family wind resistance (not stated on the P4M spec sheets checked)
Temperature	0–40 °C	DJI operating range
Rain	≤ 0.5 mm/h	no published water-resistance rating
Visibility	≥ 4.8 km (3 statute miles)	FAA 14 CFR 107.51(c), from the control station
Altitude	≤ 120 m	FAA 400 ft limit; Flight Lab setting range
Battery	finish with the batteries carried; no emergency landing	mission success and aircraft safety
Tracking	RMS lane error ≤ 2 m; coverage ≥ 95 %	at 30 m with 70 % side overlap lanes are ~7.6 m apart; 2 m uses a quarter of the overlap

Limit	Value	Why
Control margin	tilt ≤ 30°	controller limit is 35°; under 5° left means no authority for the next gust

How the weather is coupled to the mission

Forecast variable	Converted to	Enters the flight as	Effect you see
10 m wind (u, v)	speed at survey altitude (power law), direction relative to the map	mean wind added to the drone's ground velocity	more airspeed and power upwind, drift at takeoff and turns, lean into the wind
Wind uncertainty	gust σ (turbulence + ensemble spread)	random gusts (5 s) in Flight Lab; Dryden turbulence in 6-DOF	tracking error, tilt and power spikes
Temperature, pressure, humidity	air density	hover power scaling	thin, hot, humid air drains the battery faster
Temperature	battery capacity	less usable energy in the cold	more batteries needed
Radar reflectivity	rain rate	clearance rule; visibility	NOT CLEARED in rain
Wind, humidity, rain	dust risk → visibility	clearance rule	NOT CLEARED under 3 miles

The flight controller is **not** told the wind: like a real autopilot it estimates it from the difference between ground and air velocity (3 s time constant). Steady wind is learned and held off the track; wind changes, gusts, takeoff and turns push the drone off its path. The drone leans by the angle that balances acceleration and air drag (14.4° in an 8 m/s crosswind; the 6-DOF model gives 13.9°). Not coupled: wind changing across the field or within the hour, vertical wind in Flight Lab, rain and dust as forces.

Two simulators, one answer

The planner sweeps every hour and scenario with Flight Lab's fast point-mass model; the 6-DOF rigid-body model checks the recommended and selected windows in detail. The same survey in both:

Wind	Flight Lab: energy · lane error	6-DOF: energy · lane error · max tilt
calm	17.7 Wh · 0.18 m	18.0 Wh · 0.09 m · 23°
6 m/s steady	20.4 Wh · 0.18 m	21.7 Wh · 0.10 m · 27°
10 m/s steady	25.5 Wh · 0.18 m	30.3 Wh · 0.19 m · 35°
10 m/s, gusts ±2.5	26.0 Wh · 1.97 m	36.4 Wh · 1.81 m · 35°

Both agree that steady wind mainly costs energy and tilt, and gusts cause tracking error. The 6-DOF model predicts more energy in strong wind because it includes body and rotor drag explicitly; treat the planner's battery numbers as optimistic in strong wind.

Best things to do next

Ranked by how much they help the hackathon result for the time they take.

Before the event (tonight)

#	Do this	Why
1	**Run one real forecast on HiPerGator** for a real UF field and the demo date: `python earth2_weather.py --source stormcast --lat … --lon … --mission-time "…" --members 4`. Keep the output and demo from it (no live GPU needed on stage).	Turns the demo from mock to real AI weather; judges will ask.
2	**Check the hackathon rules on existing code.** Present Flight Lab as the starting platform and make the forecast, planner and physics clearly the event work.	Avoids disqualification.
3	**Rehearse a 3-minute demo** (script below) on the computer you will present from.	Live demos fail on setup, not on code.
4	**Know the limits** (see “How the decision is made”) for Q&A: what is model output, what is estimated.	Honest answers score better than big claims.

During the event (pick two or three)

Idea	Effort	Impact
Wind-aware return home. The simulator decides when to go home using calm-air power, which is why strong headwinds end in emergency landings. Make the reserve include the headwind, then show the emergency landings disappear.	Medium	High: "our simulator found a real safety bug"
Explain the decision with Gemma (MLH Best Use of Gemma): feed the hour’s reasons and numbers, get a plain-language briefing a grower can read.	Medium	High: visible AI + sponsor prize
Hindcast check: run the forecast for a past windy day and compare with what was observed (HRRR analysis or a weather station).	Medium	High credibility
Real field outline and location for a UF/IFAS plot instead of a local map.	Low	Medium
Twinly: when an hour is NOT CLEARED, let the desktop agent reschedule the calendar event and notify the team.	Low	Sponsor prize
After the flight: run a crop-health model on the captured images (your later phase). Keep for after the hackathon unless time is left.	High	Medium in 48 h

3-minute demo script

Time	Show	Say
0:00	Title, the survey map	"Growers plan drone surveys days ahead. Today they guess about the weather."
0:20	Draw survey area (2), Plan launch window (10)	"Earth-2 forecasts the field; our simulator flies the survey in every forecast scenario."
0:50	Drag the slider (12) through the day	"09:00 is CLEARED. By 11:00 blowing dust drops visibility under the FAA 3-mile limit: NOT CLEARED."
1:20	Map: green / orange / blue	"Green is calm air, orange the recommended path, blue the window you asked for."
1:40	Run 6-DOF physics (22), Play (23)	"A full rigid-body model: watch it lean into the gusts; 41 % more energy than calm air."
2:20	Battery needed, best hour	"Fly at 09:00 with one battery. That is the decision, made two days ahead."
2:40	Tests, assumptions	"Physics checked against hand calculations; estimates are labelled."

Pitch tip: lead with the decision ("fly at 09:00, not 11:00"), then show how. One strong number beats ten features.